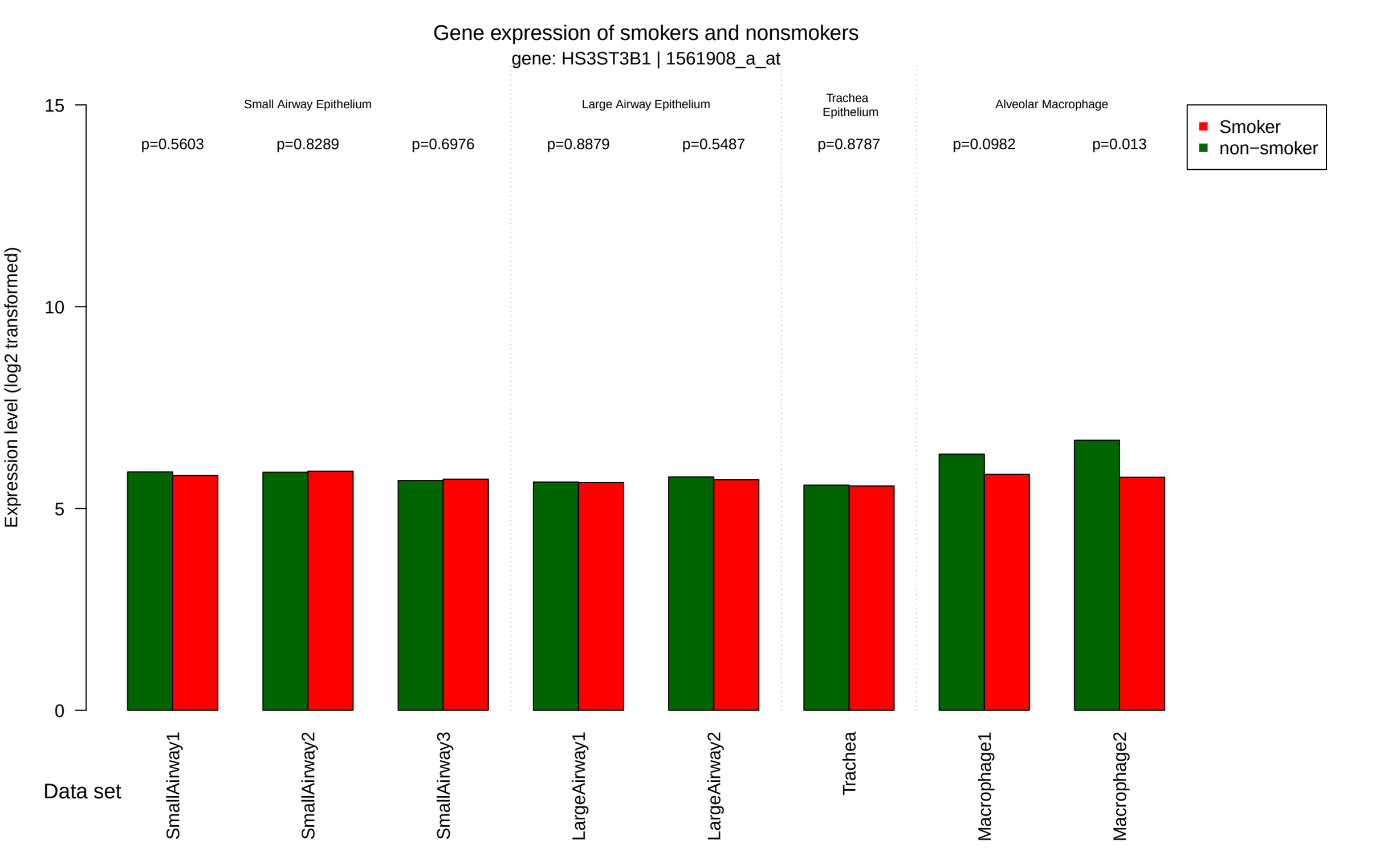
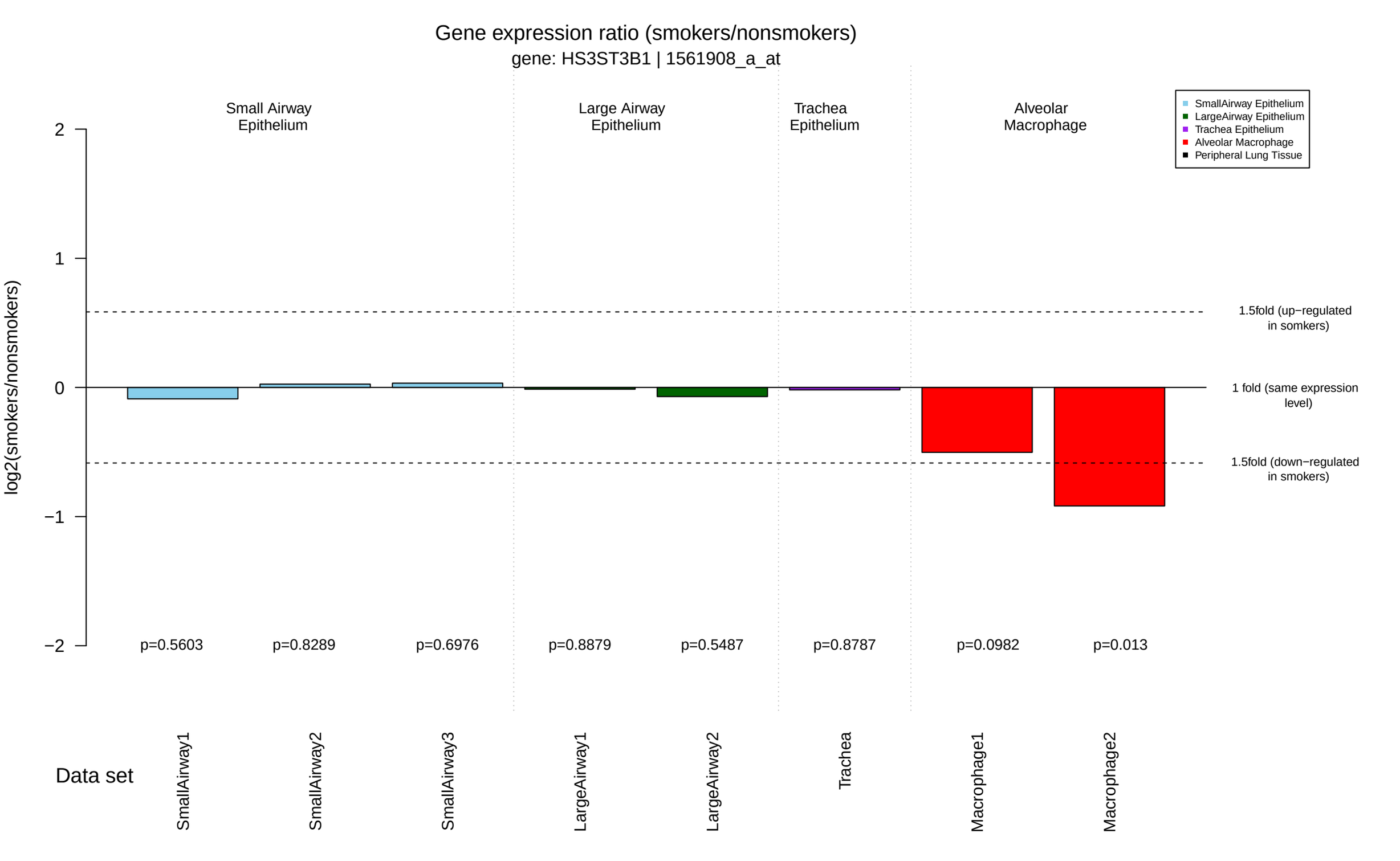
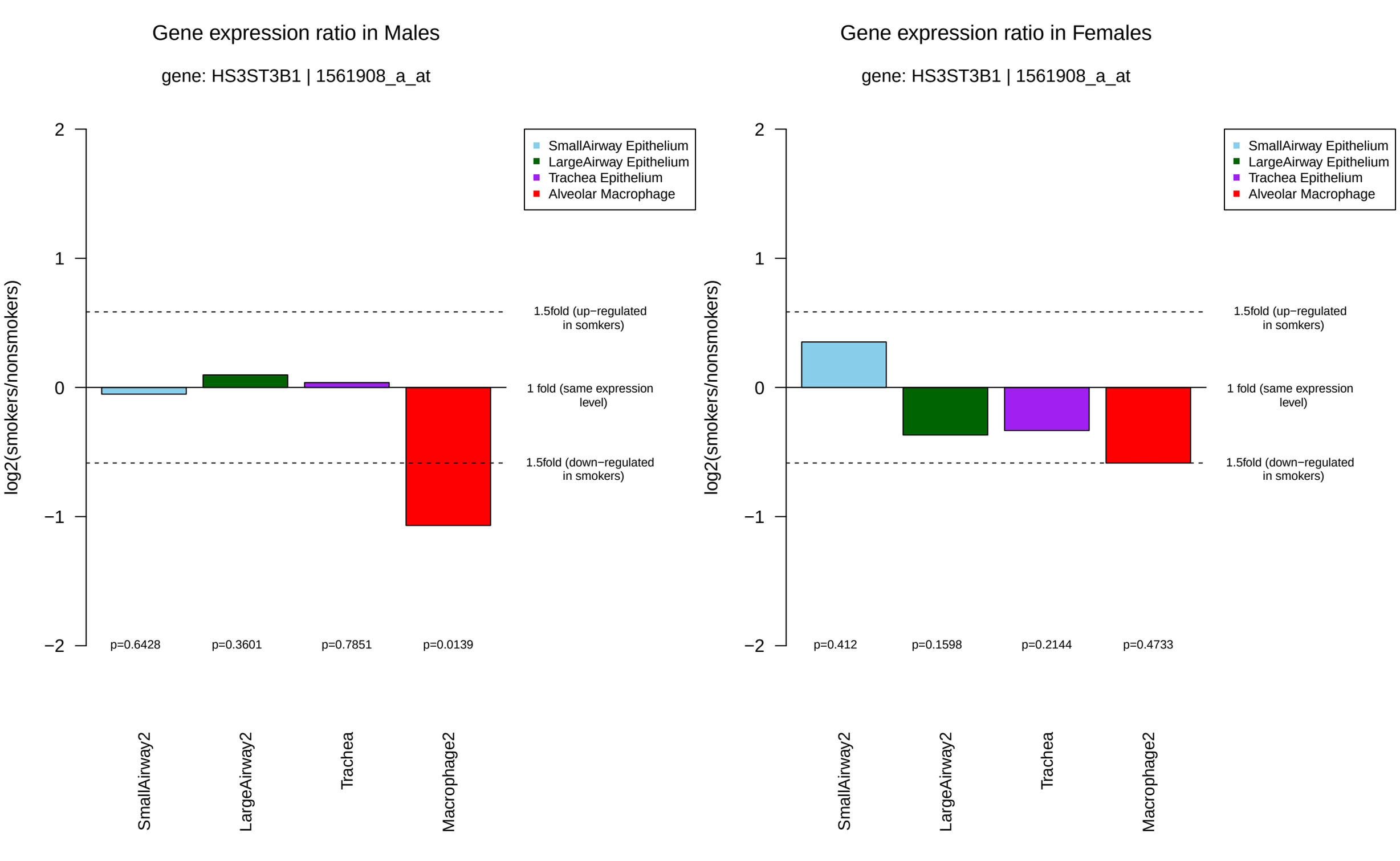
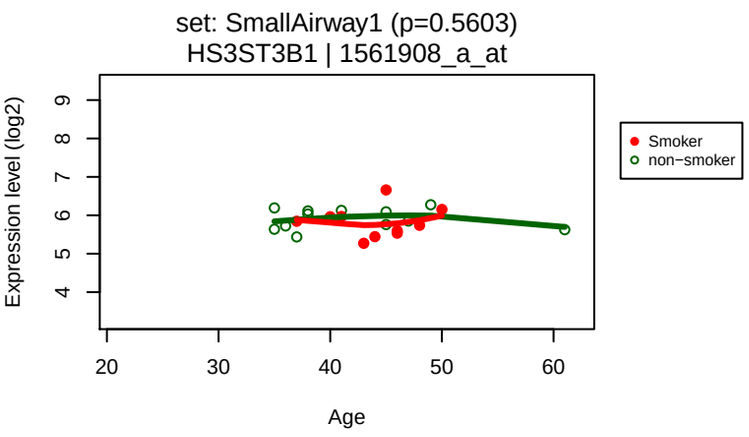
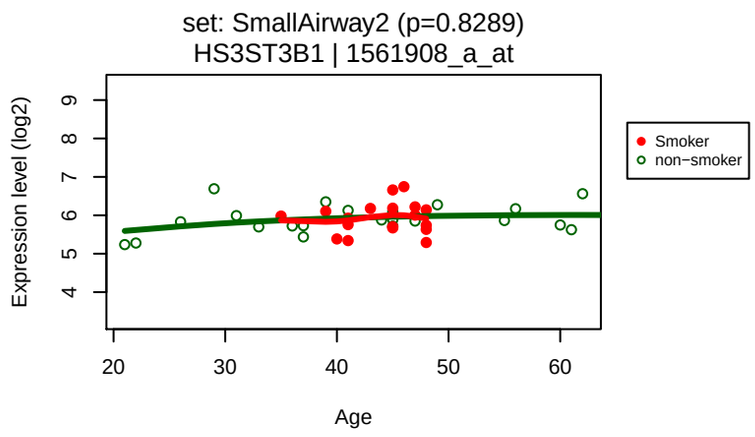
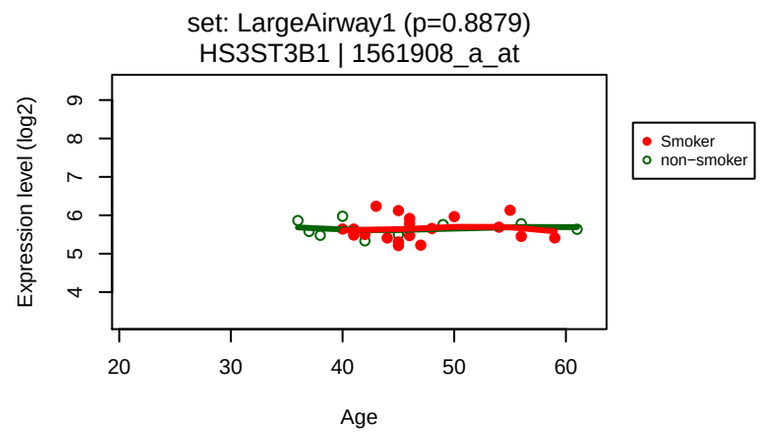

Gene expression of smokers and nonsmokers
gene: HS3ST3B1 | 1561908_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5603
p=0.8289
p=0.6976
p=0.8879
p=0.5487
p=0.8787
p=0.0982
p=0.013
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

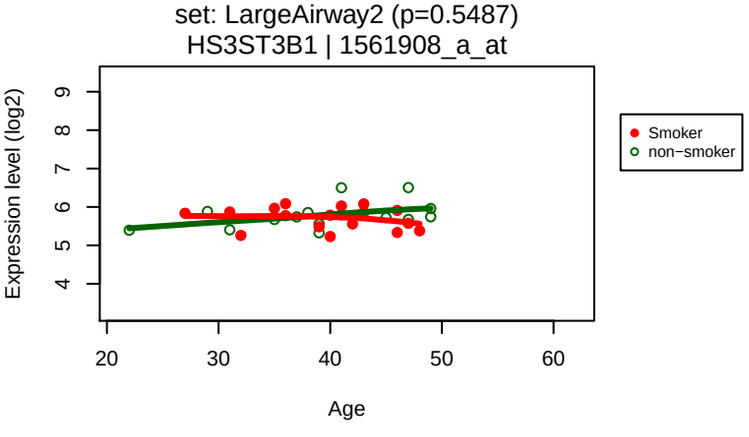

Gene expression ratio (smokers/nonsmokers)
gene: HS3ST3B1 | 1561908_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5603
p=0.8289
p=0.6976
p=0.8879
p=0.5487
p=0.8787
p=0.0982
p=0.013
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

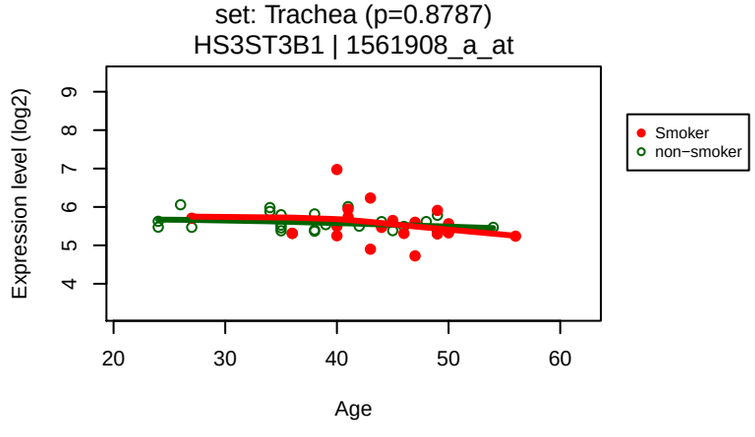

Gene expression ratio in Males
gene: HS3ST3B1 | 1561908_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6428
p=0.3601
p=0.7851
p=0.0139
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HS3ST3B1 | 1561908_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.412
p=0.1598
p=0.2144
p=0.4733
SmallAirway2
LargeAirway2
Trachea
Macrophage2

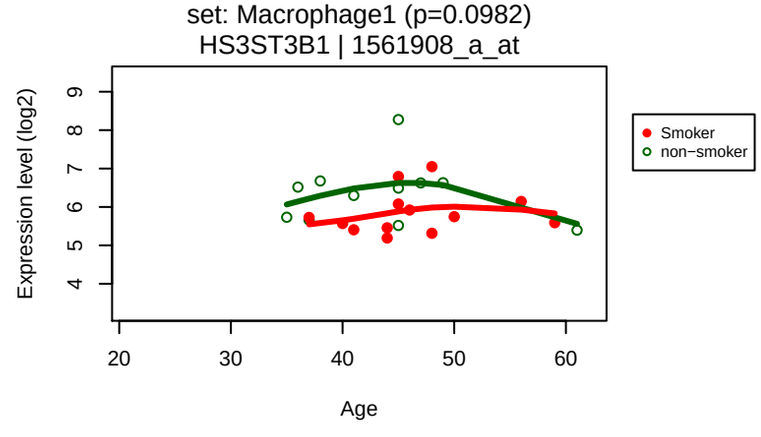

set: SmallAirway1 (p=0.5603)
HS3ST3B1 | 1561908_a_at
Expression level (log2)
Age
Smoker
non−smoker

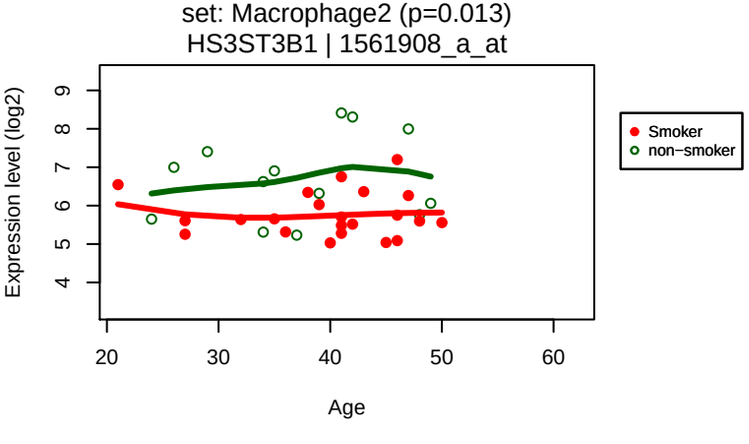

set: SmallAirway2 (p=0.8289)
HS3ST3B1 | 1561908_a_at
Expression level (log2)
Age
Smoker
non−smoker

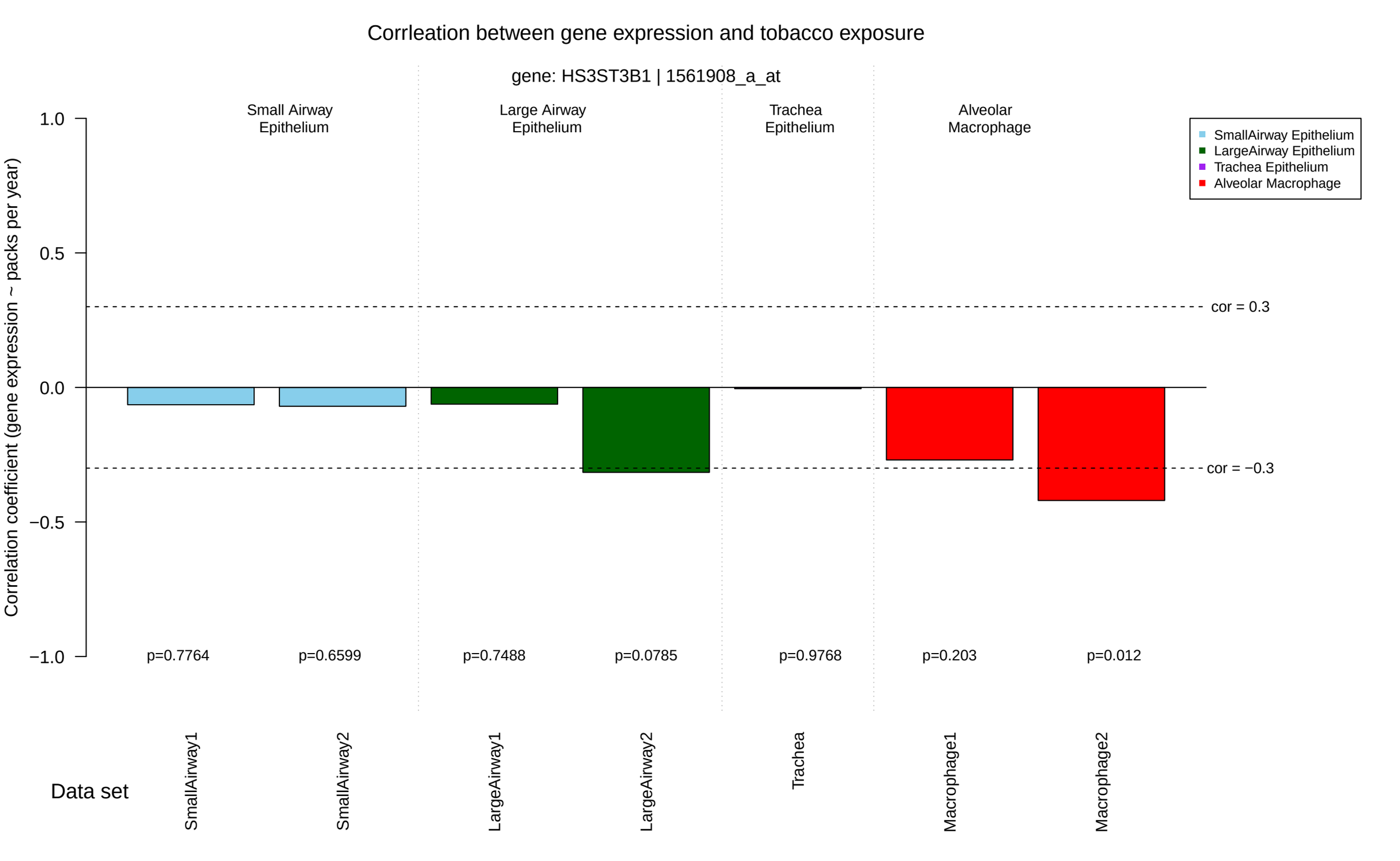

set: LargeAirway1 (p=0.8879)
HS3ST3B1 | 1561908_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5487)
HS3ST3B1 | 1561908_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8787)
HS3ST3B1 | 1561908_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0982)
HS3ST3B1 | 1561908_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.013)
HS3ST3B1 | 1561908_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HS3ST3B1 | 1561908_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7764
p=0.6599
p=0.7488
p=0.0785
p=0.9768
p=0.203
p=0.012
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2